

DataCORE

From Describing Data to Making Claims – Inference & Regression

Afternoon: DataCORE

1. Statistical Inference — Is the Difference Real?
2. Regression — Predicting One Thing from Another
3. Scientific Storytelling — Results Language

*Until now you **described** your data — means, charts, spread. Today you make the leap to a **claim about a pattern**: is what you see real, or could it just be chance?*

1:00 – 2:00 · Statistical Inference

➡ The big shift today

Describing (what you did all week):

- "Men earned \ \$27.91/hr, women \ \$25.63/hr."
- A fact about *this* sample. No argument yet.

Inferring (today):

- "Is that \ \$2.28 **gap real** in the wider world — or just luck of who we sampled?"
- A *claim* that could be **wrong**, so we test it.



Every sample wobbles. Two groups will **almost never** have identical averages by chance alone. Inference asks: **is this gap bigger than the wobble?**

Three terms, in plain English

p-value

"If there were really **no** difference, how surprising is the gap I got?" A **small** p (< 0.05) = "this would be surprising by chance, so I doubt 'no difference'."

Confidence interval

"A plausible **range** for the true value." "The gap is about \\$2.30, give or take — somewhere between \\$1 and \\$3.50." The range matters more than one number.

Statistically significant

Just means "*unlikely to be pure chance*" (small p). It does **NOT** mean big, important, or causal.

 **"Significant" $\neq$ "large" $\neq$ "important."** With enough data, a trivial \\$0.05 gap can be "significant." Always ask **how big**, not just *is it significant*.

What a p-value is – and is **not**

A p-value IS:

- The chance of a gap **this big or bigger** *if the groups were truly equal*.
- Small p → evidence **against** "no difference."

A p-value is NOT:

- The probability your hypothesis is true.
- The probability the result was "a fluke."
- A measure of **how big** the effect is.

 Convention: $p < 0.05$ = "statistically significant." It's a useful line in the sand – **not** a law of nature.

Our PBL data – the wage survey

Python R

```
wages = pd.read_csv("wages.csv")  
print(wages.head())
```

	education	gender	region	experience_years	hourly_wage
0	Bachelor	Male	Rural	10	28.54
1	Associate	Male	Urban	11	28.03
2	Graduate	Female	Suburban	4	31.56
3	Bachelor	Female	Urban	16	24.90
4	High School	Male	Urban	16	27.83

```
print("N =", len(wages), "workers")
```

N = 220 workers

 Each row is a worker: their **education**, **gender**, **region**, **years of experience**, and **hourly wage**.

First, **describe** the two groups

Python R

```
print(wages.groupby("gender")["hourly_wage"].agg(["mean", "std", "count"]).round(2))
```

	mean	std	count
gender			
Female	25.63	6.12	111
Male	27.91	5.66	109

❓ Men average more here. But the groups **overlap** a lot (look at the std). Is the gap **real**, or could two random groups differ this much? → **run the test.**

🍷🍷 RcF — a two-group t-test with `ttest_ind()`

Python (scipy)

R (base)

```
men = wages.loc[wages["gender"] == "Male", "hourly_wage"]
women = wages.loc[wages["gender"] == "Female", "hourly_wage"]

t_stat, p_val = stats.ttest_ind(men, women)          # compare two group means
print(f"difference in means: ${men.mean() - women.mean():.2f}")
```

difference in means: \$2.27

```
print(f"t-statistic: {t_stat:.2f}")
```

t-statistic: 2.86

```
print(f"p-value: {p_val:.4f}")
```

p-value: 0.0046

💡 Independent-samples t-test: **two separate groups**. R's `t.test()` even prints the **confidence interval** for the gap. (*R lists Female first, so its t shows the opposite sign — same size, same p .*)

Read the output – real, or chance?

- $p < 0.05$? → the gap is **statistically significant** – unlikely to be pure chance.
- $p > 0.05$? → we **can't rule out chance** – "no significant difference."

 Significant **does not** mean the gap is large or that *gender* caused it. Older workers earn more – if one group is more experienced, that could drive the gap. **Hold that thought for regression.**

 Sentence: "*Men earned significantly more per hour than women ($t = 2.86, p = 0.005$).*"

RcS – how **big** is it? Cohen's d

A p-value says "*probably real*." **Effect size** says "*this big*." Cohen's d = the gap measured in standard deviations.

Python

R

```
def cohens_d(a, b):
    n1, n2 = len(a), len(b)
    pooled_sd = np.sqrt(((n1-1)*a.std()**2 + (n2-1)*b.std()**2) / (n1+n2-2))
    return (a.mean() - b.mean()) / pooled_sd

print(f"Cohen's d = {cohens_d(men, women):.2f}")
```

Cohen's d = 0.39

~0.2 small

~0.5 medium

~0.8 large

💬 Report **both**: "*significant ($p < 0.05$), a small-to-medium effect ($d \approx 0.4$)*." The p earns credibility; d tells you if anyone should care.

RcS – 3+ groups? One-way ANOVA

A t-test compares **two** groups. To compare **education levels** (four groups) at once, use **ANOVA** – "are *any* of these group means different?"

Python (scipy)

R (base)

```
groups = [g["hourly_wage"].values for _, g in wages.groupby("education")]
f_stat, p_anova = stats.f_oneway(*groups)      # one-way ANOVA across all groups
print(f"F-statistic: {f_stat:.2f}")
```

F-statistic: 64.17

```
print(f"p-value:      {p_anova:.4g}")
```

p-value: 1.048e-29

```
print()
```

```
print(wages.groupby("education")["hourly_wage"].mean().round(2))
```

```
education
Associate    24.41
```

Team Task

Run the test on your data – ~12 min

1. Pick **two groups** in your dataset (or use `region:` Urban vs Rural).
2. Run the t-test — `stats.ttest_ind()` (Python) **or** `t.test()` (R).
3. Report the **difference**, the **p-value**, and **Cohen's d**.
4. **Stretch:** 3+ groups? `stats.f_oneway()` (Python) / `aov()` (R).

Write one sentence for a policy audience:

*"Group A earned \\$_ more than Group B. This difference **was** / **was not** statistically significant ($p = _$), a **small** / **medium** / **large** effect."*

12:00

2:00 – 2:45 · Regression

From "are they different?" to "how are they related?"

- A t-test compares **groups**.
- **Regression** measures how an outcome moves with a **number** — e.g. does wage rise with **experience**?
- The model draws the **best straight line** through the cloud of points.

= The line: **wage = intercept + slope × experience**

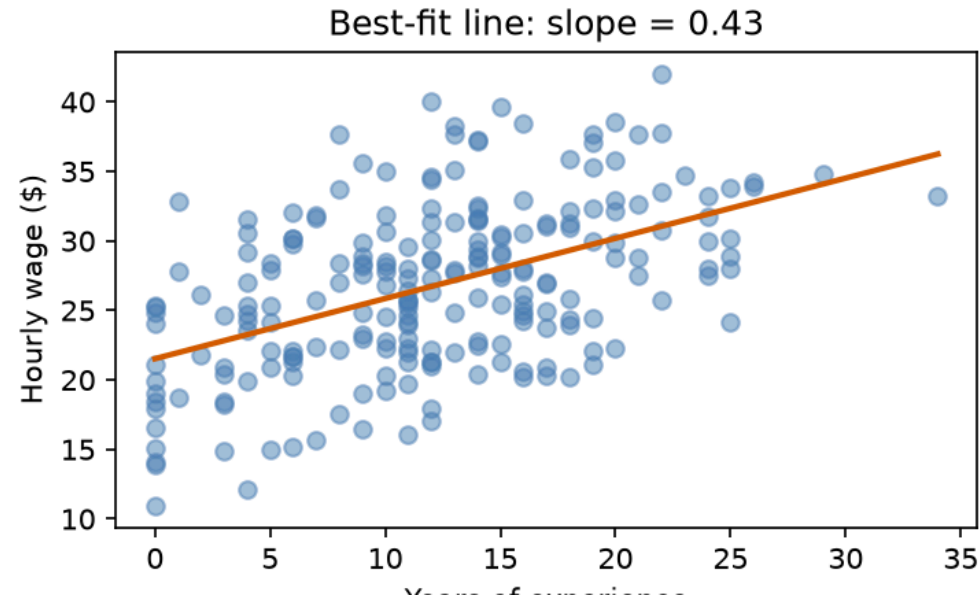
The **slope** is the headline: *how much wage changes per extra year.*

DV (outcome, y) = hourly wage · **IV** (predictor, x) = years of experience.

RcF – see the line of best fit

Python R

```
x = wages["experience_years"]; y = wages["hourly_wage"]
b1, b0 = np.polyfit(x, y, 1) # slope, intercept
plt.scatter(x, y, alpha=0.5, color="#4a7fb5")
plt.plot(np.sort(x), b0 + b1*np.sort(x), color="#D55E00", lw=2)
plt.xlabel("Years of experience"); plt.ylabel("Hourly wage ($)")
plt.title(f"Best-fit line: slope = {b1:.2f}")
plt.show()
```



RcS – OLS regression with statsmodels

Python (statsmodels)

R (base)

```
model = smf.ols("hourly_wage ~ experience_years", data=wages).fit() # DV ~ IV
print(model.summary().tables[1])
```

```
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept      21.5223       0.720     29.893     0.000     20.103     22.941
experience_years  0.4330       0.052      8.340     0.000       0.331       0.535
=====
```

 The row that matters is `experience_years` — its **coef / Estimate** is the slope. The intercept is the predicted wage at 0 years.

Reading the output table, column by column

coef (slope) = 0.433 -> wage rises \$0.43/hr per extra year

std err = 0.052 -> uncertainty around the slope

t = 8.34 -> coef divided by std err

P>|t| = 0.0000 -> is the slope distinguishable from 0?

R-squared = 0.242 -> share of wage variation explained

 **p < 0.05** on the slope = experience has a real link to wage. **R²** says how *much* of the story experience tells — here, only part of it.

Say the slope out loud

“For every **1 additional year** of experience, hourly wage increases by about **\\$0.43**, on average.”

The template:

*"For every 1-unit increase in **X**, **Y** changes by \ units, on average."*

Significant vs meaningful:

\\$0.43/yr → ~\\$4/hr over a decade. **Is that a lot?** That's the real-world question the p-value can't answer.

Significant $\neq$ meaningful – always ask both

Is it significant?

Look at **P>|t|**. Below 0.05 = unlikely to be chance.

Is it meaningful?

Look at the **coef** in real units.
\\$0.45/hr/yr – would a policymaker act on it?

How much is explained?

Look at **R²**. A real slope can still leave most variation unexplained.

 A slope can be **highly significant and tiny**, or **large but uncertain**. Report the **number with its units**, never just the stars.

Team Task

🏁 One predictor, one sentence – ~12 min

1. Pick a numeric **outcome (DV)** and numeric **predictor (IV)**.
2. Fit it — `smf.ols("DV ~ IV", ...).fit()` (Python) or `lm(DV ~ IV, ...)` (R).
3. Find the **coef**, its **p-value**, and **R²**.
4. Plot the scatter + best-fit line.

Write the interpretation sentence:

*"For every 1-unit increase in **[IV]**, **[DV]** changes by **[coef]**, on average ($p = \text{---}$, $R^2 = \text{---}$)."*

Then discuss: **is the magnitude meaningful**, not just significant?

12:00

2:45 – 3:00 · Scientific Storytelling

Two registers – you need **both**

The number (for credibility)

"The regression coefficient on experience was 0.45 (SE = 0.06, $p < 0.001$, $R^2 = 0.21$)."

Other researchers can **check** this.

The plain language (for the audience)

"Workers earned about 45 cents more per hour for each additional year of experience."

A policymaker can **act** on this.

 Strong results writing puts the **plain sentence first**, then backs it with the numbers in parentheses. Never make the reader decode statistics to learn what you found.

Draft your team's headline finding – now

Write **one results sentence** for your main finding, in this shape:

*"Our analysis found that **[plain-English finding]** (**[stat]** = , **p** =)."*

Then read it to a partner who is **not** a statistician. Did they get it?

✓ Checklist:

- Plain-English claim **first**?
- Direction clear (more / less / higher)?
- The number **with units**?
- p-value (and effect size / R^2) in parentheses?
- No causal verb unless you can defend it?

10:00

Real-World PBL Case

The climate question – real public data

- **NOAA**: U.S. summer temperature, by year.
- **NIFC**: total acres burned by wildfire, by year.
- **Research question**: is a **hotter summer** associated with **more acres burned**?
- 30 years of data: **1993–2022**.

 One row per **year** – this is **time-series** data, not people. The same tools (correlation, regression) still apply.

Python R

```
clim = pd.read_csv("climate_wildfire.csv")
print(clim.head())
```

	year	summer_temp_anomaly_F	acres_burned	acres_burned_millions
0	1993	0.6	1800000	1.80
1	1994	-0.3	4070000	4.07
2	1995	0.5	1840000	1.84
3	1996	-0.1	6070000	6.07
4	1997	0.4	2860000	2.86

Plot the 30-year trend – both rising?

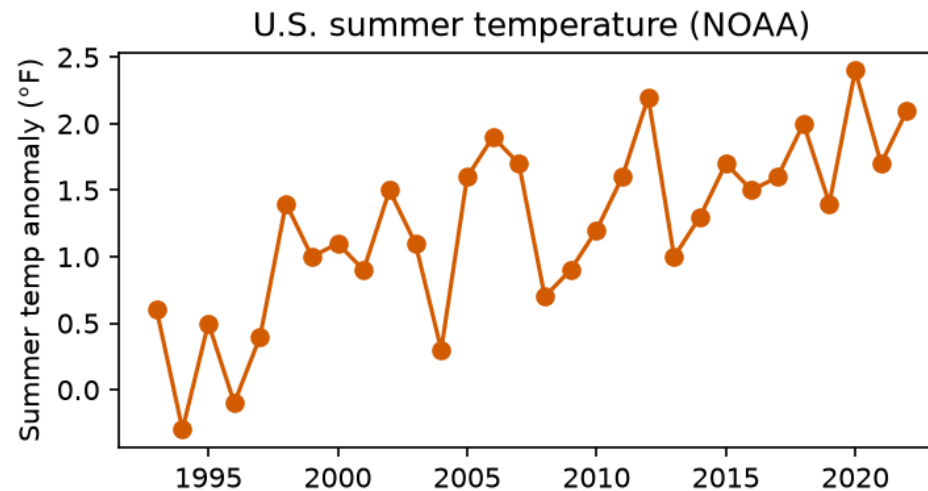
Temp · Python

Temp · R

Acres · Python

Acres · R

```
plt.plot(clim["year"], clim["summer_temp_anomaly_F"], marker="o", color="#D55E00")
plt.xlabel("Year"); plt.ylabel("Summer temp anomaly (°F)")
plt.title("U.S. summer temperature (NOAA)")
plt.show()
```



👁 Both **drift upward** over 30 years — but year-to-year both are noisy. Eyeballing isn't enough; **measure** the association.

Correlation – do they move together?

Python (scipy)

R (base)

```
r, p = stats.pearsonr(clim["summer_temp_anomaly_F"], clim["acres_burned_millions"])  
print(f"Pearson r = {r:.3f}")
```

Pearson r = 0.604

```
print(f"p-value = {p:.4f}")
```

p-value = 0.0004

r runs from **-1 to +1**.

0 = no link · +1 = perfect rise-together · here **r** $\approx$ **0.60** = a **moderate positive** association.

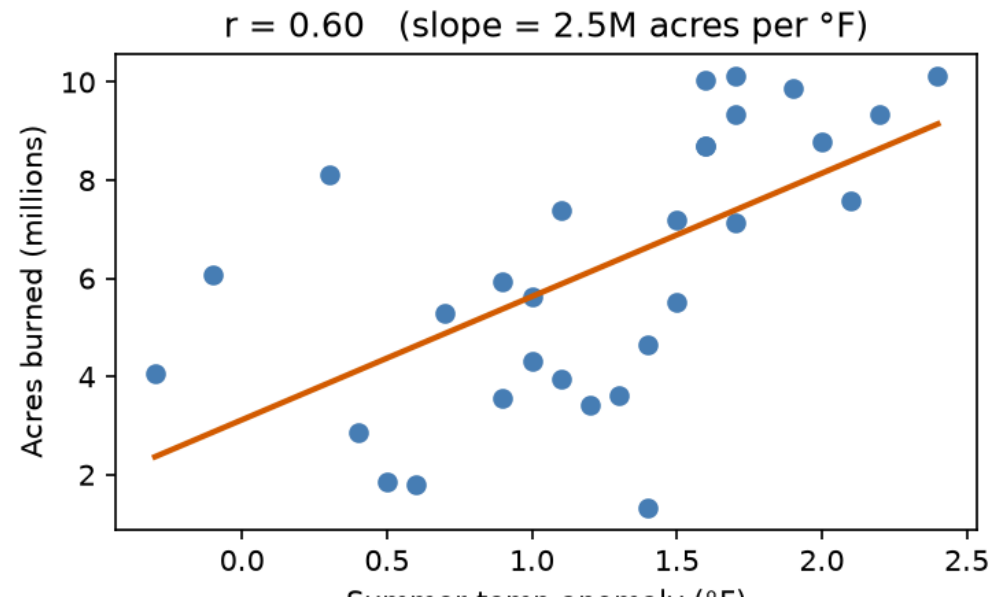
p < 0.05 → this correlation is **unlikely to be chance**.

Hotter summers **do** tend to come with more acres burned.

■ ■ ■ The scatter – temperature vs acres burned

Python R

```
x = clim["summer_temp_anomaly_F"]; y = clim["acres_burned_millions"]
m, b = np.polyfit(x, y, 1)
plt.scatter(x, y, color="#4a7fb5")
plt.plot(np.sort(x), b + m*np.sort(x), color="#D55E00", lw=2)
plt.xlabel("Summer temp anomaly (°F)"); plt.ylabel("Acres burned (millions)")
plt.title(f"r = {r:.2f} (slope = {m:.1f}M acres per °F)")
plt.show()
```



⚠ Is this correlation **causal**? Not so fast.

🚫 **Correlation is not causation.** A real, significant correlation does **not** prove that heat *causes* the acres burned. Other factors move with both.

Drought

Dry years are hot **and** burn more — drought may drive both.

Forest management

Decades of fire suppression left more fuel to burn.

Land use

More homes in fire-prone areas → more ignitions & more reported acres.

🔍 These are **confounders** — lurking variables that could explain the link. Naming them honestly is what separates analysis from a hot take.

Write the honest climate result

“ Across 1993–2022, hotter U.S. summers were **associated with** more acres burned ($r \approx 0.60$, $p < 0.05$). This is a **correlation, not proof of cause** – drought, forest management, and land-use change are plausible confounders that future analysis should account for.”

☑ Notice the careful verbs: "**associated with**," not "*caused*." The number gives credibility; the caveat keeps you **honest**.

